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Studies on the Cs_3Sb Photo-Cathode

By TOSHIMICHI SAKATA

The Electrical Communication Laboratory

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Spectral and energy distributions for external photoelectrons ejected from Cs_3Sb photo-cathodes were determined by the method of interchangeable emitters. Precautions were taken to secure electrically uniform surfaces.

The energy distribution had an usual form for $h\nu=2.14, 2.27, 2.84, 3.06, 3.39$ and 3.96 ev. The results indicated that Cs_3Sb were semiconductors with a small values of $\delta=0.2\sim0.3$ ev. Here δ is the energy difference between the Fermi level and the top of the occupied band of energy states. The photoelectric work function (ϕ') was found about 1.8 ± 0.1 ev. The work function obtained from the contact potential techniques were about $1.5\sim1.6$ ev. On the other hand, the thermionic work function was found about $1.3\sim1.5$ ev, using Richardson's formula. Typical values of $m^{(2)}$ [parameter depending both on the form of occupied band and on the energy dependence of the photo-electric excitation probability] obtained both from energy and spectral measurements were 1. Thermoelectric measurements showed that all samples were P type.

§ 1. Introduction

It has been known for many years that photoelectric layers of Cs_3Sb are extremely sensitive photoelectrically. At optimum sensitized condition, these layers can be con-

sidered as an intermetallic compound Cs_3Sb .

The detailed electronic state of the substance, however, are not clear at present. The surface photoeffect in metals has so far been calculated by using Sommerfeld's assumption

of completely free electrons within the metal. In the case of semiconductors, however, it is assumed that the energy bands are completely filled and the free electron theory is no longer adequate. Following Apker and his collaborators¹⁾, we suppose that the number of electrons $N(E) dE$ emitted with a given energy E by light is proportional to the numbers of electrons with an equivalent energy ϵ within the material and proportional to the transition probability P . When the electrons emitted come from states in the upper edge of a filled band, we can use the fact that the density of states near the upper edge is known to be const. $\times (\epsilon_0 - \epsilon)^{1/2} d\epsilon$, where ϵ_0 is the highest energy in the band. They further assumed that the probability P is proportional to the velocity of the "hole" created by the removal of the electrons.

The foregoing discussion is useful for discussing energy and spectral data for Cs_3Sb photo-cathodes. In the work reported here, photoelectric emission, contact potential and thermoelectric measurements have been made on several Cs_3Sb photo-cathodes, in order to obtain further information concerning the energy level structure.

§ 2. Fundamental Formulas

Simplifying assumptions made in reference¹⁾ led to the following expression for the current-voltage characteristic at 0°K

$$(I/I_s) Y = \int_E^{E_m'} N(\nu, E) dE \propto q(\nu) (h\nu - \varphi')^{m+2} \\ \times [(m+2)(1-\nu')^{m+1} - (m+1)(1-\nu')^{m+2}] \quad (1)$$

where

$$\nu' = (V_s - V)/(V_s - V_0')$$

E : Kinetic energy of the photoelectron

E'_m : the maximum energies from a semiconductor

m : parameter depending both on the form of occupied band and on the energy dependence of the photoelectric excitation probability.

q : slowly varying function of ν

Y : the photoelectric yield

S : the probability of excitation from the state ϵ to the state $(\epsilon + h\nu)$.

If S is proportional to the velocity of the hole formed in the filled band by the transition, $m=1$.

The value of m can be found from the

plots which best fit the experimental curves. An independent value may be obtained from the spectral distribution.

§ 3. Experimental Details

Fig. 1 shows the construction of the phototubes used in this work. A is the emitter support (Mo wire 0.7 mm in diameter) with interchangeable emitters (B_1, B_2), hanging on it, and D is the quartz window that

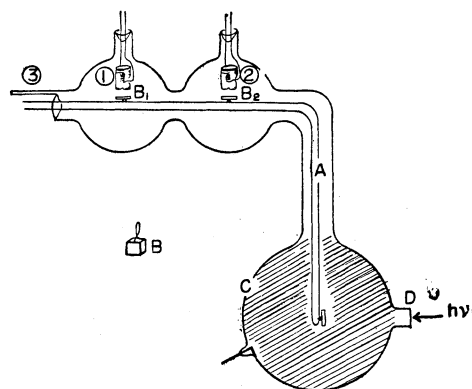


Fig. 1. Typical experimental tube for measuring contact potential and energy distribution of the photoelectrons.

admits the radiation. The inside of the glass sphere is coated with a conducting layer of Ag, which serves as the collector. The glass spheres were 60 mm in diameter and these emitters $4 \times 5 \text{ mm}^2$ in diameter. Two evaporation chambers were mounted in a side tube. The thin films of Sn were formed onto the Mo plate (B_2) in Fig. 1 by evaporating Sn from the tungsten spiral (2).

The thin films of Cs_3Sb were formed on the B_1 (Mo plate) in another chamber. The first step in the preparation of Cs_3Sb is to produce a layer of Sb. The next step in the process is the introduction of Cs into the phototube through (3) in Fig. 1.

After the baking process is completed, the completed phototube is sealed off from the pumping system. It is also well known fact that the Cs introduced in the tube in the second process above mentioned escapes to all directions and condenses on the colder parts of the glass wall or else upon a conducting metallic coating such as Sn previously deposited. Such a metallic emitter is named "Sn-Cs" layer. Spectral sensitivity

of the Sn—Cs layer is surprisingly different from that of the Sn in the region of visible ray (see Fig. 4.)

But it is evident that this Sn—Cs layer shows metallic characters. [See section 4 (A)]

The collector C, (Ag films), is found to be similarly photosensitive in the visible region, although pure Ag metal is known to respond only to ultraviolet radiation. This fact may be correctly attributed to the deposition of Cs atoms on the collector.

It may be possible to evaporate the adsorbed Cs atoms on the glass wall (Collector) by heating. But these heating processes were avoided, because it may occur, by heating of Cs, the deposition of excess Cs atoms on the surface of the Cs_3Sb photo-cathodes. Radiation sources for energy measurements were high pressure Hg discharges (250 watt).

The desired spectral lines were isolated with a quartz monochromator. Intensities were constant within 0.3%. On the emitter, the area of spot of light were $0.8\sim 1.2\text{ mm}^2$.

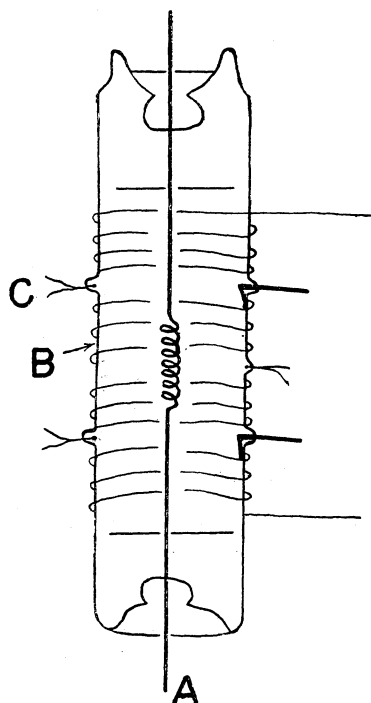


Fig. 2. Experimental tube for the measurement of conductivity and thermionic emission.

For spectral measurements, a tungsten (standard) lamp a high pressure Hg discharge lamp were used as the light source. Inten-

sities were determined with a $(Sb+B_i)-(Sb)$ thermopile (2.8Ω , 3 mm^2).

Currents were measured with a UX-54A electrometer tube with a $5\times 10^{10}\Omega$ grid resistor and a sensitivity of 1 mm per 0.1 mV. Next, thermoelectric measurements on Cs_3Sb were done, and these results were compared with the foregoing results based on the photoelectric measurement. All of the experiments were made in tubes with a central anode and two conductivity strips of Mo, encompassing the photo-cathodes. For this purpose tubes as shown in Fig. 2 were adopted, in which A and B were plate and cathode respectively. C is Cu-constantan thermocouple 0.03 mm in diameter welded in the glass bulb. By mounting three thermocouple in the cathode, they were able to make simultaneous measurements of the conductivity and thermionic emission of the cathode. Thus, precaution were taken to observe the exact value of temperature of the cathode. The cathode is heated by means of direct current in the coil placed around the tube (diameter 27 cm and 125 cm long).

The temperature of cathodes is variable from room temperature up to 120°C with accuracy $\pm 0.5^\circ\text{C}$. After considerable experimenting, the following technique of measurements (conductivity and thermionic emission) was developed :

- (1) Heater power was raised to an amount sufficient to maintain the temperature 371°K . During the measurements, three thermocouple indicated the same value of mV.
- (2) Heater power was then reduced to zero, and hence the cathode was cooled to room temperature (At least forty minutes was allowed for cooling).
- (3) The measurement of thermionic emission were performed with the well known method, using the UX-54A amplifier.

The thermo-electromotive force of Cs_3Sb was measured using the UX-54A amplifier. The construction of experimental tube was quite same as described in the previous paper.³⁾⁻⁴⁾

§ 4. Discussion of Results

(A) Metallic Emitter

In Fig. 3 are shown the current-voltage curves for the Sn—Cs metal at room temperature for the $h\nu=2.84$, 3.06 and 3.39 ev.

In this case the Sn-Cs metallic emitters were used to determine the stopping potential V_0 at 0°K by Du Bridge's method.

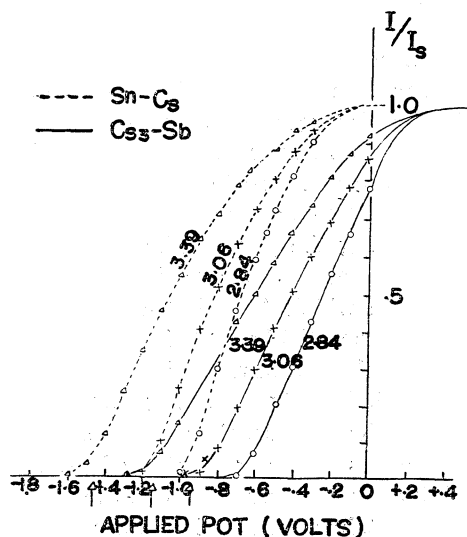


Fig. 3. Current-voltage characteristics for Cs_3Sb and for a metallic emitter Sn-Cs. The arrow marks V_0 , "the stopping potential for 0°K." Currents are normalized at $V=3$ volts. To avoid confusion, only a few of the experimental points are given.

This reference point, $V_0=(\varphi_c-h\nu)/e$, corresponding to the coincident Fermi levels of the emitters, is marked by an arrow for various values of $h\nu$. Here φ_c is the work function of the collector. The values of V_0 are shown in Table I.

Table I.

$h\nu$ (ev)	V_0 (v)	φ_c (ev)
2.84	-0.95	1.90
3.06	-1.15	1.91
3.39	-1.49	1.89

Fig. 4 shows the current-voltage curves plotted on a parabolic scale. Currents were normalized at $V=3.0$ volt, a point definitely on the saturation line.

Saturation points determined by this extrapolation process were $V_s=0$. The value of $\phi[h\nu-e(V_s-V_0)]$ determined from Einstein relation, was 1.90 ± 0.01 (ev). Fig. 5 shows the spectral distribution of photoelectron from these representative Sn-Cs emitters. In Fig. 5, one notices that the photoelectrons emerge in two groups. The first may be called the Cs-group, since these

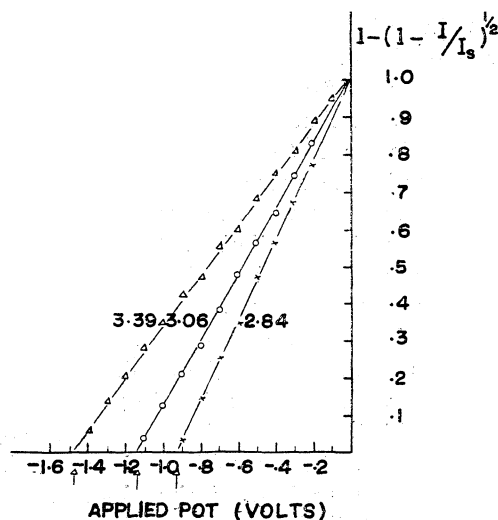


Fig. 4. Current-voltage data for 2.84, 3.06 and 3.39 eV. plotted on a parabolic scale. The lower arrows show values of V_0 determined from Du Bridge's analysis.

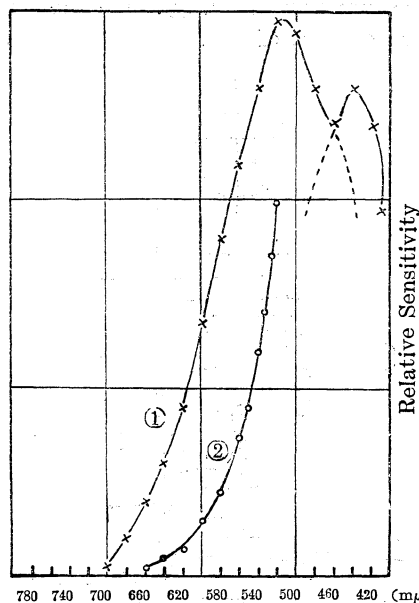


Fig. 5. Photoelectric emission from Sn-Cs. In Curve 1 the corresponding photo-currents ($\times$) are plotted for the light source. In curve 2 the photo-currents ($\circ$) are shown as calculated for an ideal light source radiating equal amounts of energy at every wavelength.

photoelectrons behave as though ejected in normal fashion from the Cs film, whereas the second may be termed Sn-group. Fig. 6 shows the Fowler plot for normal incidence for $T=300^\circ\text{K}$. The work function determined from Fig. 6 was

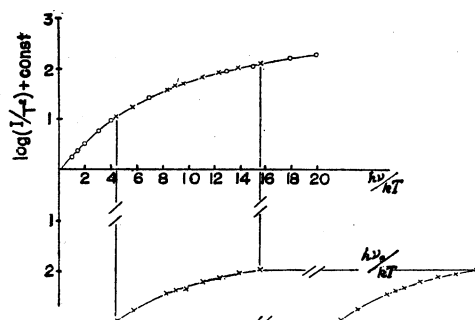


Fig. 6. Fowler plot for normal incidence
for $T = 300^\circ\text{K}$

$$\phi = 1.87 \pm 0.03 \text{ (ev) .}$$

The results above mentioned show that the methods of Fowler and Du Bridge give consistent values for the work function of Sn—Cs film. Thus it is evident that $\phi = \phi_c = 1.90$ ev. Other values of $h\nu = 3.96$ ev gave the same result for ϕ_c .

(B) Cs_3Sb emitter

Fig. 3 shows Current-voltage characteristics for Cs_3Sb and for a typical metal Sn—Cs having the work function, $\phi = 1.90$ ev. The curves for Cs_3Sb had a saturation point V_s , which determines the condition of zero field between emitter and collector, is indicated at $+0.4$ volt. Current-voltage characteristics failed to show clear-cut saturation points. One way to minimize this difficulty is to illuminate the emitter with low light intensity for several wavelengths. ($h\nu = 2.84, 3.06$ and 3.39 electron-volt.) Fig. 7 shows the lower portions of characteristics analogous to those in Fig. 3, except that the ordinate scale is expanded by a factor of 10.

At these levels a measurable fraction of the photo-current may be due to stray radiation scattered from the emitter to the collector and then to the emitter support. Hence the curves (after correction for reverse currents) represent upper limit to the current from the Cs_3Sb . The curve for Cs_3Sb approaches the axis tangentially and falls to unmeasurable values at point $V_0 \sim -1.2, -0.9$ and -0.7 volt respectively for $h\nu = 3.39, 3.06$ and 2.84 ev. Thus as compared with Sn—Cs metals of the work function $\phi = 1.9$ ev, the Cs_3Sb emitted very few electrons having energies within a range $\delta = e(V_0' - V) \sim 0.3$ ev of the Einstein maximum $h\nu - \phi$. This range

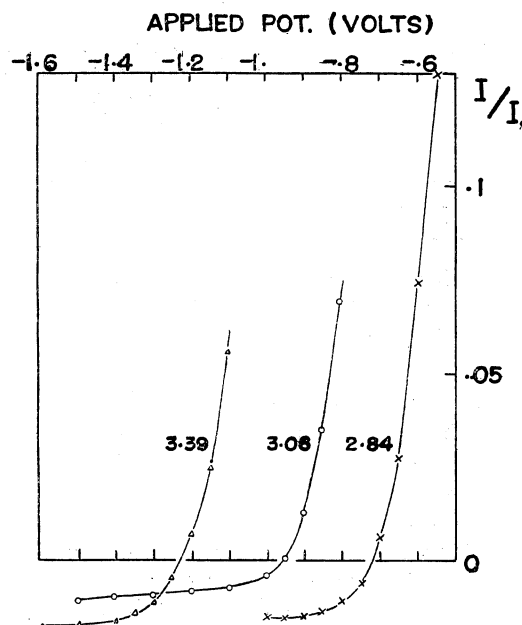


Fig. 7. Lower portions of curves like those in Fig. 3 (Cs_3Sb). The ordinates for $h\nu = 2.84, 3.06$ and 3.39 ev are expanded by a factor of ten in the inset to show the behaviour of the exponential temperature "Tails" in the presence of reverse currents.

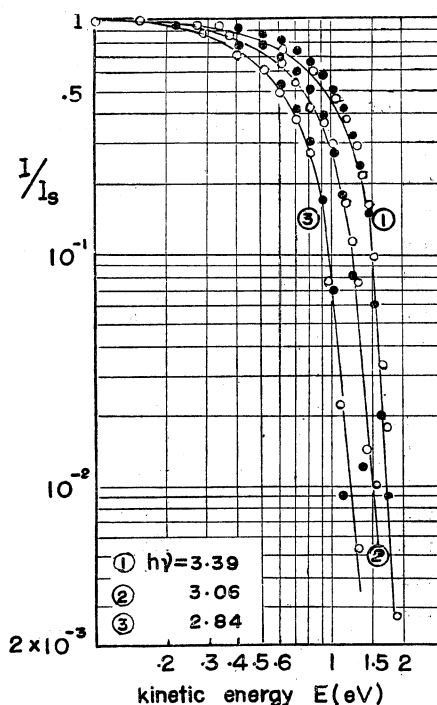
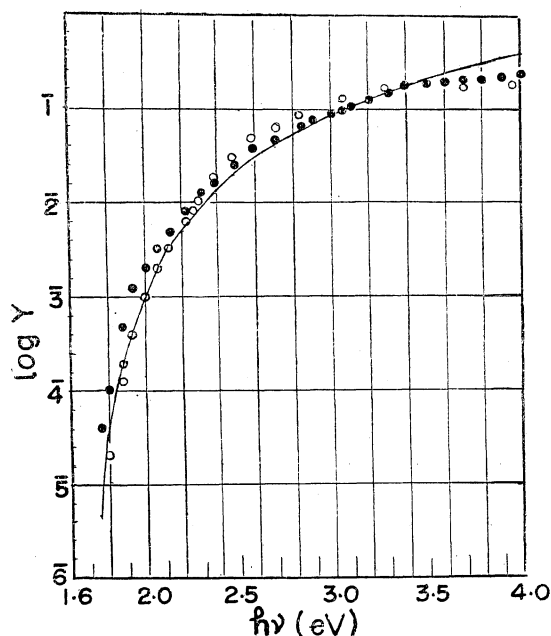


Fig. 8. In this figure, open and filled symbols show logarithmic plots of current vs. kinetic energy for Cs_3Sb at room temperature. The solid line is computed from the Equation (1) using $m = 1, \delta \sim 0.3$ ev.
○: Tube No. 1; ●: Tube No. 2.

Fig. 9. Spectral distribution of the yield from the Cs_3Sb . The smooth solid line is calculated from the relation $Y = (1/2.8) \times 10^{-1} (h\nu - 1.7)^3$. Filled symbols are the results reported for Cs_3Sb by Apker, Taft and Dickey⁵⁾. The open symbols shows the magnitude of the observed value.



was correlated with the forbidden energy interval lying between Fermi level μ and the top of the occupied band of Cs_3Sb .

Another tests made at $h\nu = 3.96, 2.27$ and 2.14 eV gave the same results. Fig. 8 shows the logarithmic plots of current *vs.* kinetic energy for Cs_3Sb .

The solid curves are calculated from Eq. (1) with $m=1$. When $m=1$ and $\delta=0.30$, this relation leads to characteristics like those observed for Cs_3Sb .

Fig. 9 shows the spectral distribution of the photoelectric yield Y near the threshold for Cs_3Sb . Here, the value of Y at $h\nu = 3.39$ eV are adjusted to coincide with Apker et al's⁵⁾. The spectral distribution of the yield can be described approximately by the relation $Y = C \times 10^{-1} (h\nu - \phi')^3$ (electrons/quantum),

Table II. Photoelectric yield (Y) from the Cs_3Sb .

$h\nu$	Apker Taft and Dickey (Log Y)	(Calc.) $Y = C \times 10^{-1} (h\nu - \phi')^3$ (Log Y)	(Obs.) (Log Y)
ev	electrons/quantum		
4	1.4	1.62	1.25
3.9	1.3 ₈	1.58	—
3.8	1.3 ₃	1.52	—
3.7	1.3 ₁	1.45	1.23
3.6	3.39	1.38	1.23
3.5	3.3	1.32	—
	3.2	1.08	—
	3.06	2.93	1.10
2.84	3.0	2.89	—
2.7	2.9	2.78	—
2.58	2.6	2.38	2.70
2.48	2.4	2.23	2.52
2.38	2.2 ₇	2.04	2.24
2.29	2.1	3.88	2.02
2.27	2.21	3.83	3.96
	2.13	3.70	3.80
	2.06	3.46	3.54
	2.0	3.26	3.31
	1.93	4.98	3.00
	1.87	4.56	4.60
	1.8	4.23	4.13
	1.76	5.60	5.32
		6.85	

where $h\nu < 3.60$ ev. Here ϕ' is the photo-electric threshold energy and C is a constant dependent on samples. Thus both the energy and spectral distributions are indicative of the same energy structure.

By the above experimental formula the quantum yield Y is calculated for every wave-length, of which values are shown in Table II.

(C) Thermoelectric measurements

The saturated thermionic emission from the photo-cathode was determined as a function of temperature, and Richardson plots

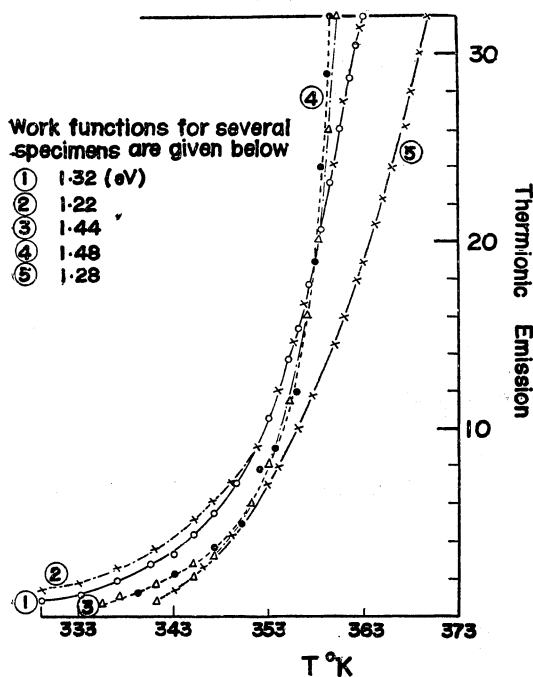


Fig. 10. Thermionic emission vs. temperature curves for Cs_3Sb photo-cathodes.

Table III.

ϕ (ev)	$\Delta E/2$ (ev)
1.2 ₀	0.30
1.2 ₂	0.28
1.3 ₂	0.26
1.4 ₄	0.27
1.4 ₈	0.25

were made. These results are shown in Fig. 10 and Fig. 11. The thermionic work function determined from the slope of these curves for various samples ranged from 1.2₂ to 1.4₈ ev. The activation energies $\Delta E/2$ of their

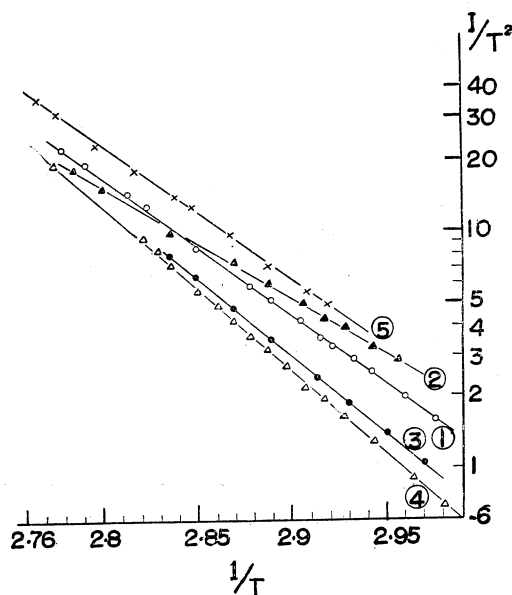


Fig. 11. Richardson plots for Cs_3Sb photo-cathodes.

conductivities measured at the same time respectively. These values were shown in Table III. As previously³⁾⁻⁴⁾ indicated the sign of thermo-electromotive force showed the P-type semiconductor for Cs_3Sb photo-cathodes. The relation between ΔT and $\Delta\theta$ (thermal electromotive force) were linear over the range of ΔT (temperature difference) from 0° to 15°C. The typical results are shown in Fig. 12. From the gradients of these

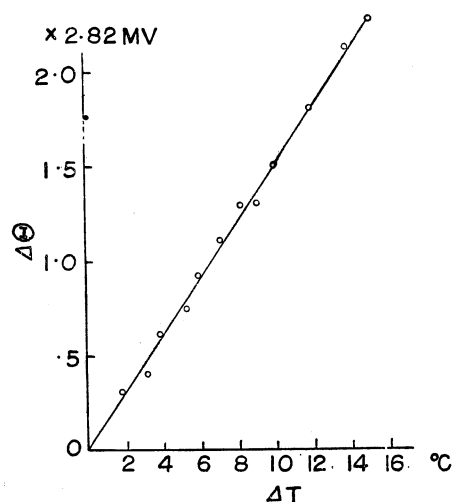


Fig. 12. The relation between the T.E.M.F $\Delta\theta$ and any temperature difference ΔT .

straight lines, $d\theta/dT$ can be obtained. The warmer electrode was always negative, in-

dicating positive charge carriers. The activation energy $\Delta E/2$ of the conductivity simultaneously were 0.24 and 0.26 eV.

Thermo-electromotive force at room temperature has values 0.42~0.44 eV/deg. The chemical potential thus had values of 0.18~0.19 eV. The atomic concentration n_b of acceptors was estimated at room temperatures, these values were always $5 \times 10^{19} \sim 10^{20}$ per cm^3 . The relation between the number n_b of impurity atoms in unit volume and the number n of electron holes in unit volume is

$$n = n_b^{1/2} [2 \cdot (2\pi m k T)^{3/2} / h^3]^{1/2} \exp(-\Delta E/2kT),$$

where ΔE is the energy separation between the top of filled band and the impurity level.

The values of n were consequently $4.8 \times 10^{15} \sim 10^{16}$ per cm^3 .

§ 5. Summary

The following points are suggested by the experimental results.

(1) In spherical phototubes with interchangeable emitters, the energy and the spectral distributions of Cs_3Sb are indicative of the same energy structure ($m=1$, $\delta \sim 0.3$ eV). It was very difficult to find definite stopping potentials corresponding to upper edges of occupied band in Cs_3Sb .

The work function thus found for Cs_3Sb was 1.4~1.5 eV. Within the experimental

uncertainties the Cs_3Sb was a semiconductor with $\delta \sim 0.3$ eV.

(2) The spectral distribution of the quantum yield near threshold can be described approximately by the relation

$$Y = C \times 10^{-1} (h\nu - \phi')^3,$$

consistent with the values $m=1$. Here ϕ' is the photoelectric threshold and C is constant depending on specimens.

(3) The thermionic emission and conductivity results appear to be capable of interpretation by a semiconductor picture as a P type semiconductor.

(4) The sign of the thermoelectromotive force indicates positively charged carriers in agreement with these above concept.

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